Project guide

Football Transfer Market

How the project was planned, built, verified and deployed: the full stack explained for learning, with diagrams, algorithms, problems hit and how they were avoided.

Live site: https://frontend-theta-three-tj1p5tsbe7.vercel.app

Repository: github.com/ssupreethsg/football-transfer-market (branch `claude/data-pipeline-v2`)

Scope: 2,707 players · 96 clubs · Premier League, La Liga, Serie A, Bundesliga, Ligue 1 · season 2026/27

Written: 31 October 2026

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1 The project in one page

What it is. A website that lists every player at every club in Europe's top five leagues and shows, for each player, real statistics for his whole career: totals per competition (league, domestic cups, European competitions, national team) and a season-by-season breakdown where every season is split into each competition he played.

How it works, in one sentence. A Python pipeline downloads real match data, turns it into clean JSON files, checks them against an independent source, and a Next.js (TypeScript/React) website hosted on Vercel reads those files and displays them.

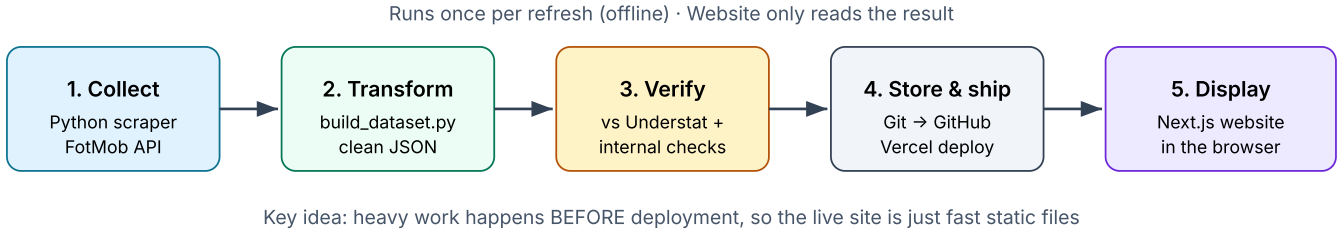


Figure 1. The five steps of the system.

Key numbers

Item	Value
Clubs / players	96 clubs, 2,707 players (current 2026/27 squads)
Requests made to collect the data	~28,400 (5 league tables, 96 squads, 2,707 profiles, 25,628 detailed-stat entries, 1 league list)
Detailed season-competition entries	23,693 (minutes, xG, dribbles, tackles… for 2023/24 onwards)
Dataset size shipped to the site	~57 MB in 2,709 JSON files (1 index, 1 meta, 2,707 player files)
Independent verification	League apps/goals/assists agree with Understat for 99.1% / 99.5% / 98.9% of 4,489 player-seasons
Live-site test	2,707/2,707 player files return HTTP 200; 150 pages clicked through in a real browser with 0 errors
Code written	~780 lines Python (pipeline), ~660 lines TypeScript/TSX (website)

2 The starting plan

Before writing code we answered four questions. This is the framework to reuse for any project: **What must it do → What limits us → What shape should the system take → What tools fit that shape.**

2.1 Requirements (what you asked for)

1. Real, correct data, collected from the internet, not estimated.
2. Every player at every club in the top 5 leagues.
3. For each player: **cumulative career stats** split by competition, and **yearly stats** where each year is split by competition (league, Carabao/EFL Cup, FA Cup, Champions League, Super Cup, national team...).
4. Goal contributions and deeper stats: goals, assists, dribbles, and more.
5. Verify the data again after collecting it.
6. Website must work for anyone at any time, even with your laptop off; no errors.
7. Save everything in the `football-transfer-market` repository.

2.2 Constraints (what limits us)

Constraint	Consequence for the design
Most football stats sites block bots (Cloudflare "Just a moment..." pages, HTTP 403/405).	Pick a source that serves JSON to normal requests; don't fight anti-bot systems.
No source publishes "official" data for all 5 leagues + all cups + Europe in one place.	Use one consistent aggregator (Opta data via FotMob) and cross-check it with an independent one (Understat).
~2,700 players × many seasons × many competitions = tens of thousands of requests.	Cache every response on disk, run in parallel with a rate limit, make it resumable.
Site must work with your laptop off, free hosting, no server to maintain.	Pre-compute everything; serve static JSON files from Vercel's CDN. No live backend.
Limited tokens / time.	Build one example player end to end first, then run the same code for everyone.

2.3 The design decision: a "static data site"

There are two common shapes for a data website:

> **Option A: live backend (what the first version tried)**
> Browser → API server (FastAPI) → database (SQLite/Postgres). Needs a server running 24/7, costs money, breaks when the server/laptop is off; the first version even pointed at `127.0.0.1:8000` (your laptop).

> **Option B: pre-built static data (what we chose)**
> Pipeline builds JSON files once → files are deployed with the website → browser downloads only the file it needs. No server to crash, free, very fast, works with the laptop off. Trade-off: data updates when you re-run the pipeline.

Player stats change at most once per match day, so "rebuild and redeploy" is a perfect fit. This is the single most important architectural choice in the project.

2.4 The plan as a checklist

#	Step	Tool / language	Output
1	Find root cause of the wrong data	Reading old Python code	Found "projected 2024 → 2026" numbers
2	Test which data sources are reachable and complete	curl, Python	FotMob (primary), Understat (verification)
3	Write a cached, resumable scraper	Python + requests + threads	`cache/` raw JSON
4	Transform raw data into the shape the site needs	Python	`public/data/*.json`
5	Build one example (Haaland), inspect, fix issues	Python + screenshots	Validated structure
6	Rewrite the website to read the new data	TypeScript, React, Next.js, Tailwind	Home + player pages
7	Run for all players, verify against Understat	Python	`verification_report.json`
8	Browser-test pages, fix every error found	Playwright (Node.js)	0 errors
9	Commit, push, deploy preview, test, promote to production	Git, GitHub, Vercel CLI	Live site

3 Architecture: the big picture

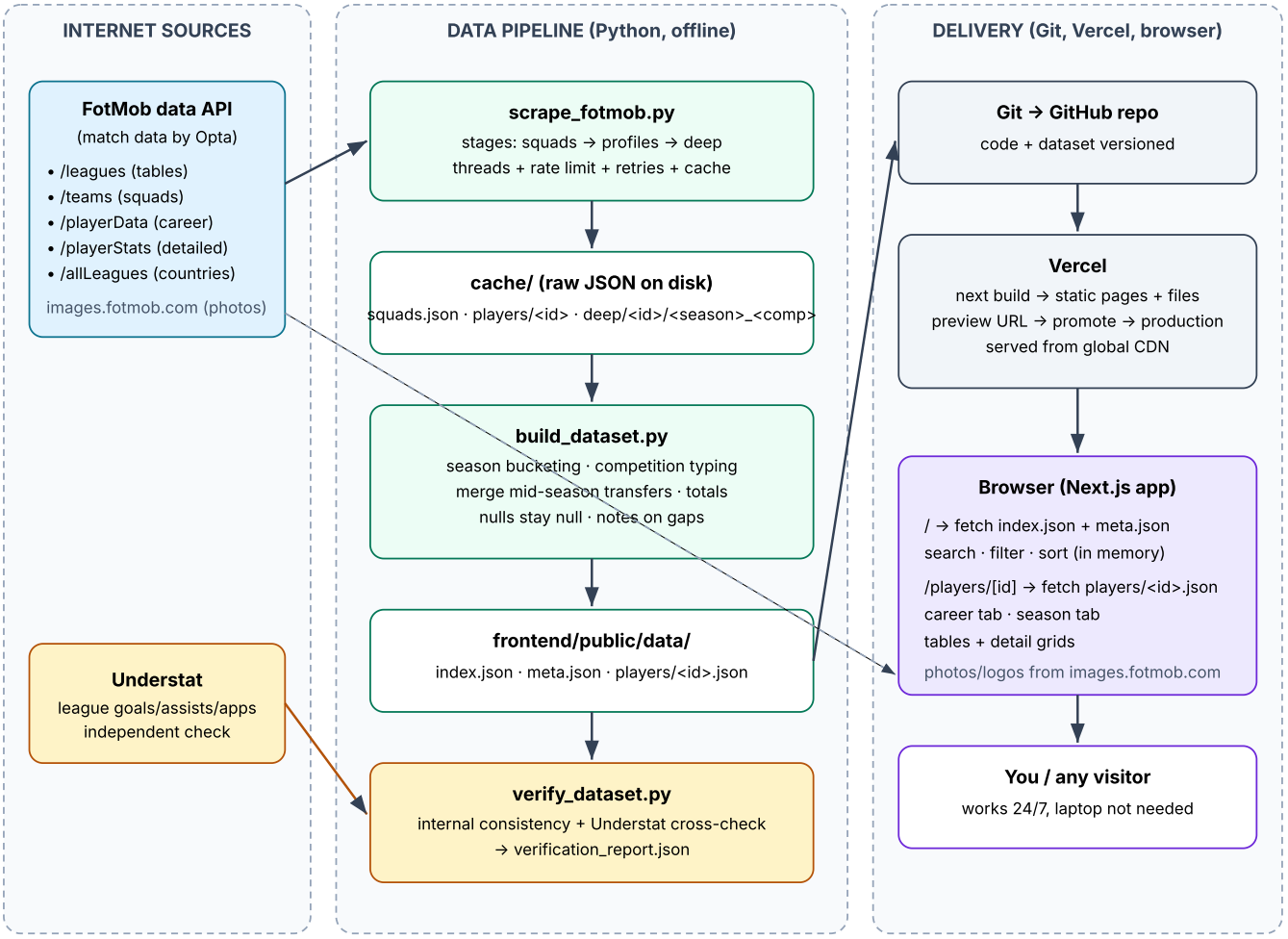


Figure 2. System architecture. Left: where data comes from. Middle: the offline pipeline. Right: how it reaches visitors.

Read the diagram like this

Everything in the middle column runs on a computer (this time, a cloud container) when you want fresh data. Its only output is a folder of JSON files. Those files are committed to Git and deployed with the website. When a visitor opens the site, no Python runs at all: the browser downloads small JSON files and React draws the tables.

4 Technology stack

4.1 Languages

Language	Where	Why this language
Python 3.13	Data pipeline: `scrape_fotmob.py`, `build_dataset.py`, `verify_dataset.py`	Best ecosystem for data work; simple HTTP + JSON; readable scripts.
TypeScript 5	Website logic: `src/lib/*.ts`, pages	JavaScript with types: catches mistakes like a misspelled field before the site runs.
TSX (React JSX)	Website UI: `page.tsx`, `PlayerView.tsx`, `Img.tsx`	Write HTML-like markup inside TypeScript; React turns it into the page.
CSS via Tailwind 4	Styling through class names like `bg-slate-900 rounded-2xl`	Style directly in markup; consistent spacing/colours; no separate CSS files to maintain.
JSON	The dataset and all configs	The common format both Python and the browser read natively.
Bash / shell	Running stages, git, Vercel CLI	Glue between tools.
JavaScript (Node.js)	Playwright browser tests (testing only)	Playwright's primary API.

4.2 Libraries, frameworks and services

Tool	Version	Role
requests (Python)	2.34	HTTP client for the scraper and verifier.
concurrent.futures / threading (Python stdlib)	built-in	Run up to 10 downloads at once; lock-based global rate limiter.
difflib, unicodedata (Python stdlib)	built-in	Fuzzy name matching and accent folding ("Mbappé" = "mbappe") in verification.
Next.js	16.3.3	React framework: routing (`/`, `/players/[id]`), building, optimisation.
React	19.2.8	UI library: components, state (`useState`), effects (`useEffect`), memoisation (`useMemo`).
Tailwind CSS	4	Utility-class styling.
ESLint + eslint-config-next	9 / 16.3.3	Code-quality rules (caught a React anti-pattern, see §14).
Geist font (next/font)	-	Self-hosted web font, no layout shift.
Playwright	1.56	Headless Chromium to screenshot and click through pages, record JS errors.
Git + GitHub	-	Version control and remote storage of code + data.
Vercel (+ Vercel CLI 64.5)	-	Build and host the site; preview and production deployments; global CDN.

4.3 Project file map (the parts that run the live site)

```
football-transfer-market/
└── project-material/
    ├── backend/pipeline/              ← DATA PIPELINE (Python)
    │   ├── scrape_fotmob.py           collect: squads → profiles → deep stats (cached)
    │   ├── build_dataset.py           transform: raw cache → website JSON
    │   ├── verify_dataset.py          verify: internal checks + Understat cross-check
    │   ├── verification_report.json   latest verification results
    │   ├── README.md                  how to run the pipeline
    │   └── cache/                     raw downloads (git-ignored, ~110 MB)
    └── frontend/                      ← WEBSITE (TypeScript / Next.js)
        ├── public/data/               the dataset the site reads
        │   ├── index.json             one row per player (1.2 MB)
        │   ├── meta.json              build time, season, counts
        │   └── players/<id>.json      2,707 full player files
        └── src/
            ├── app/layout.tsx         HTML shell, font, page title
            ├── app/globals.css        Tailwind import + dark theme colours
            ├── app/page.tsx           home: search, filters, player cards
            ├── app/players/[id]/page.tsx       route wrapper (server component)
            ├── app/players/[id]/PlayerView.tsx player page (client component)
            └── lib/
                ├── types.ts           TypeScript shape of the dataset (the "contract")
                ├── format.ts          number/money formatting, labels, stat groups
                └── Img.tsx            image that hides itself if missing
```

5 Where we started: what was wrong

The first version (built with another agent) looked complete but the numbers were unreliable. Reading the code found the root causes. **Lesson: when data "looks wrong", read the code that produced it before collecting new data.**

Problem found	Evidence	Effect on the site
Projected, not real, stats	`fetch_api_football.py` : "free tier only allows up to 2024. We will fetch 2024 and mathematically project it to 2026", ages +2	Every 2026 number was an estimate.
Invented contract data	Release clauses, wages (default €35,000/week), buy-back clauses had no source	Financial figures were fiction.
Stale squads	Club list hard-coded for 2024/25 (Ipswich, Leicester, Southampton…)	Wrong clubs/leagues for many players.
Fake "sync" button	`/api/sync` returned "success … synchronized" whenever the backend was unreachable	Claimed updates that never happened.
Backend on your laptop	`BACKEND_URL` default `http://127.0.0.1:8000`	On Vercel this always failed silently and fell back to bundled data.
Unknowns turned into zeros	Schema defaults like `assists = Column(Integer, default=0)`	"0 assists" displayed where the truth was "unknown".
Only top scorers/assisters fetched	Endpoints `topscorers` , `topassists`	Most players had no real stats at all.
One 14 MB file loaded by every API call	`import rawPlayers from "@/data/players.json"`	Slow functions, hard to scale.

The general lesson

A system that silently "falls back" or "fills in" values is the most dangerous kind: it never shows an error, so you don't notice the data is wrong. Our rebuild follows one rule everywhere: **if the source doesn't have a number, store `null` and show "–"; never invent, never default to 0.**

6 Data sources

6.1 How we chose: a reachability test

We sent one test request to every candidate source from the build machine and looked at the response code.

Source	Result	Decision
FBref	403, Cloudflare "Just a moment..." challenge	rejected anti-bot wall
Transfermarkt	405 (bot blocked)	rejected
Sofascore API	403	rejected
API-Football	403 without key; free tier limited to old seasons	rejected (root of the old problem)
Premier League official API	Requires pretending to be premierleague.com (fake Origin/Referer headers)	rejected blocked as bot-check circumvention; also PL-only
FotMob data API	200 JSON; full careers; every competition; detailed stats; Opta data	primary source
Understat	200 JSON; all players per league season	verification source

Ethics and terms of use

We only used sources that answer normal requests, sent requests at a polite rate, cached everything so each URL is fetched once, and never bypassed bot protection. FotMob's and Understat's terms may still restrict re-publishing their data on a public site; for a commercial product you would license data (e.g. from Opta/Stats Perform).

6.2 The FotMob endpoints we use

Endpoint	Calls	What we take from it
`/api/data/leagues?id=47` (PL), 87, 55, 54, 53	5	League table → the 20/18 club ids in each league; current season name.
`/api/data/teams?id=<club>`	96	Squad: every player id, name, position group.
`/api/data/playerData?id=<player>`	2,707	Bio (age, height, foot, nationality, contract), market value, **careerHistory**: every season × every competition with apps/goals/assists/rating, for club, national team and youth; **statSeasons**: list of season/competition pairs that have detailed stats.
`/api/data/playerStats?playerId=&seasonId=<entry>`	25,628	~50 detailed stats for one season-competition: minutes, xG, xA, shots, dribbles, chances, passes, duels, tackles, cards, goalkeeping...
`/api/data/allLeagues`	1	League → country, used to label "Bundesliga (Austria)" vs "Bundesliga (Germany)".

How we discovered these: open the public website, look at what JSON the page itself loads, then test the URL patterns with curl and inspect the keys with a short Python script. This "inspect before you build" step saved hours.

7 Pipeline stage 1: collecting the data

File: `backend/pipeline/scrape_fotmob.py` (Python). It has three stages you run in order:

```
python scrape_fotmob.py squads                       # 5 leagues → 96 clubs → 2,707 player
ids
python scrape_fotmob.py profiles --workers 10        # 2,707 career profiles
python scrape_fotmob.py deep --workers 10 --seasons 4 # 25,628 detailed-stat entries
python scrape_fotmob.py deep --player 737066          # any stage, one player only (the
Haaland example)
```

7.1 The core algorithm: a cached, rate-limited, retrying fetch

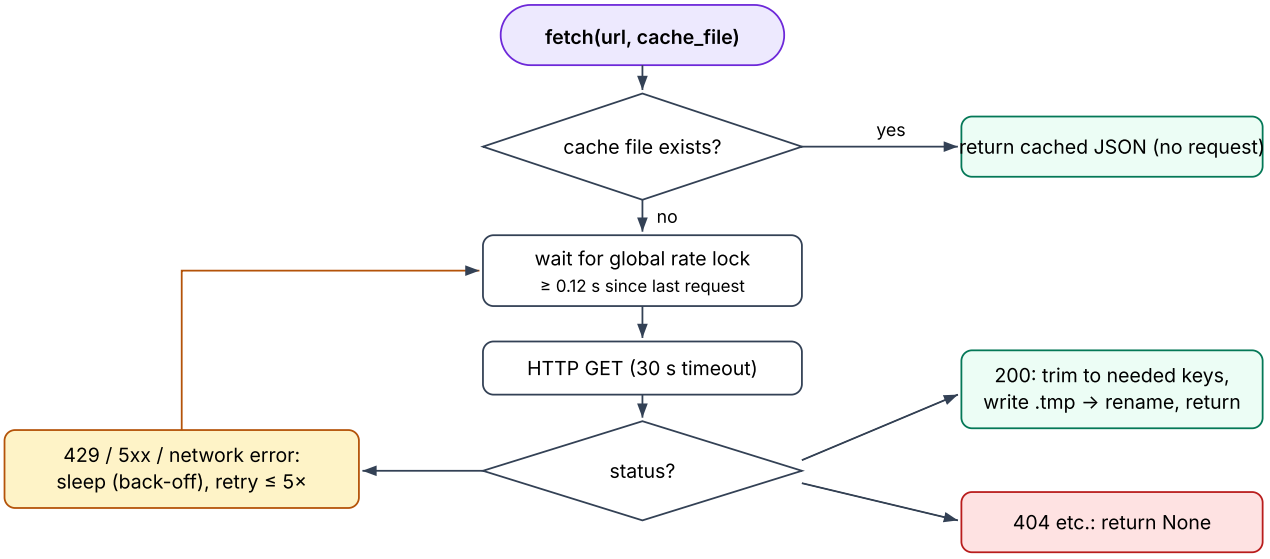


Figure 3. The fetch() function, the heart of the scraper. Every one of the ~28,400 requests goes through it.

7.2 Why each piece exists

Technique	Problem it prevents
Disk cache (one JSON file per request)	Re-downloading after a crash; hammering the source. A stopped run simply continues; a second run is instant.
Atomic write (write `.tmp`, then `os.replace`)	A half-written file if the process dies mid-write, which would otherwise be treated as valid cache.
Thread pool (10 workers)	Waiting on network latency one request at a time. First run did 0.7 profiles/s; with 10 workers ~8 requests/s (25,628 entries in ~50 min).
Global rate lock	Workers bursting together; keeps total rate polite no matter how many threads.
Retry with back-off on 429/5xx	Losing data to temporary server hiccups.
Trimming responses to needed keys	Disk usage: raw profile ~140 KB, deep stats ~130 KB (shot maps, heat maps). Trimmed cache is ~110 MB instead of ~4 GB.
Null-safe trimming	The run crashed at 25,603/25,628 because FotMob answered `null` for a few entries. Now an empty answer is stored as "no stats".
`--player` option	Testing on one player (Haaland) before spending ~1 hour on everyone.

7.3 The real code (simplified)

```
def fetch(path, params, cache_file, trim=None):
    if os.path.exists(cache_file):                # 1. cached? done
        return json.load(open(cache_file))
    for attempt in range(5):
        with _rate_lock:                          # 2. one request at a time leaves the
gate
            wait = _last_call[0] + MIN_INTERVAL - time.time()
            if wait > 0: time.sleep(wait)
            _last_call[0] = time.time()
        r = session.get(f"{BASE}/{path}", params=params, timeout=30)
        if r.status_code == 200:
            data = trim(r.json()) if trim else r.json()
            write_atomically(cache_file, data)    # 3. .tmp then os.replace
            return data
        if r.status_code in (429, 500, 502, 503, 504):
            time.sleep(5 * (attempt + 1)); continue  # 4. back off and retry
        return None                               # 5. 404: nothing there
```

7.4 Which detailed stats are fetched

The profile's `statSeasons` lists, for each season, the competitions with detailed data and an `entryId` like `"1-3"` (season #1, competition #3). The `deep` stage takes the 4 most recent seasons and every competition flagged `hasDeepStats`, and saves each as `cache/deep/<player>/<season>_<competitionId>.json`. Older seasons are skipped to keep the run to about an hour; apps/goals/assists for *all* seasons already come from the profile.

8 Pipeline stage 2: transforming the data

File: `backend/pipeline/build_dataset.py`. Input: the cache. Output: `frontend/public/data/`. This is where most of the "thinking" lives, because raw data is messy in ways you only discover by looking at it.

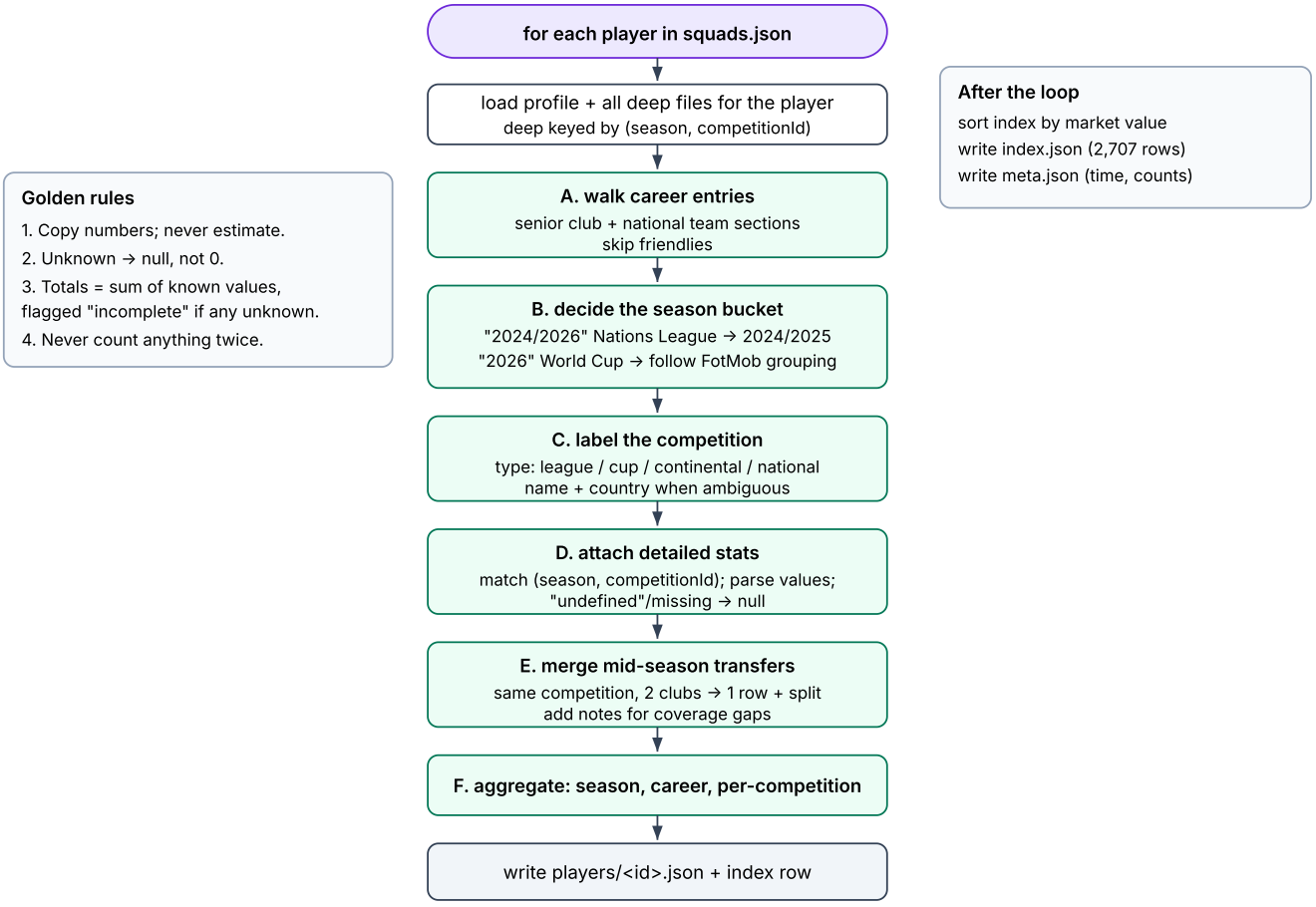


Figure 4. build_dataset.py, per player.

8.1 Step B: season bucketing (the trickiest rule)

Club seasons are named `2025/2026`, but other entries are named differently:

Raw season name	Example	Bucket we use	Why
`2025/2026`	Premier League	2025/2026	Normal club season.
`2025` inside a club entry	FIFA Club World Cup 2025	2025/2026 (the club entry's season)	FotMob already files it under the club season.
`2026` (national team)	World Cup 2026	2026/2027	Follow the season in which FotMob lists that tournament's detailed stats, so basic and detailed numbers land together.
`2024/2026` (national team)	Nations League 2024/25	2024/2025	FotMob labels some editions over two years; they are played in the first club season. Without this rule we showed fake seasons like "2020/2022".
`2018` (club)	Norwegian league (Molde)	2018	Calendar-year leagues stay as they are.

8.2 Step C: labelling competitions

- **Type** by name patterns: continental (*Champions League, Europa, Conference, UEFA Super Cup, Club World Cup…*), domestic cup (*Cup, Copa, Coppa, Pokal, Coupe, Shield, Supercopa, …mesterskap…*), national team (by section), otherwise league.
- **Name** from FotMob's league list. If the same name exists in several countries ("Bundesliga", "Premier League", "Super Cup", "Cup") the country is appended: *Bundesliga (Austria)* vs *Bundesliga (Germany)*.

8.3 Step E: merging mid-season transfers

Dean Huijsen, 2023/24: FotMob's career lists *Juventus, Serie A, 1 app* and *Roma, Serie A, 13 apps*, but the detailed Serie A stats cover the whole season (14 apps). Attaching those detailed stats to both rows counted them twice. Fix: rows with the same competition in the same season merge into one row, *"Roma / Juventus: 14 apps, 2 goals"*, with a per-club split kept underneath. This halved the career-vs-detailed disagreements and matches Understat (which reports the combined 14 apps, 2 goals).

8.4 Step F: how totals are calculated

Kind of stat	Examples	How it is combined
Counts	goals, assists, minutes, shots, dribbles, tackles	Sum of known values.
Rates and percentages	rating, pass accuracy %, dribble success %, save %	Minutes-weighted average (a 90-minute game counts more than a 5-minute cameo). Never summed.
Unknown parts	Old seasons where assists = "undefined"	Total = sum of known values, field listed in `incomplete`; the site shows a * next to it.

```
# the "unknown is not zero" rule, from build_dataset.py
def num(v, as_float=False):
    if v is None or v == "" or v == "undefined":
        return None                       # unknown stays unknown
    f = float(str(v).replace(",", "").replace("%", ""))
    return f if as_float else int(round(f))
```

8.5 Notes added to rows

- *"Detailed stats cover 11 of 16 matches (qualifying / early rounds have no detailed data)"*: e.g. Europa League including qualifiers. The headline apps/goals/assists still include qualifiers.
- *"Match data shows 1 goal… the source's season summary differs"*: the two FotMob endpoints disagree for the same matches (usually the current season, before one side updates). We show both rather than silently picking one.

9 The data model

The website reads three kinds of files. The same shape is written down twice: produced by Python (`build_dataset.py`) and described in TypeScript (`src/lib/types.ts`). That TypeScript file is the **contract** between pipeline and site: if they disagree, TypeScript reports an error at build time.

players/737066.json (one player)

```
{ id, name, team, team_id, league,
position, nationality, age, height_cm,
preferred_foot, contract_end,
market_value: {value_eur, date, source},
career_totals: {
appearances, goals, assists, incomplete[],
detailed: {minutes, xg, dribbles, …},
detailed_seasons: ["2026/2027", …] },
career_by_competition: [
{ competition, type, seasons,
appearances, goals, assists,
detailed:{…} }, … ],
seasons: [
{ season: "2025/2026", teams: [..],
totals: {apps, goals, assists},
totals_detailed: {…},
competitions: [
{ competition, type, team, edition,
appearances, goals, assists, rating,
detailed: {~50 stats} | null,
by_team: [...] | null, notes: [] }
] }, … ],
source: {provider, url} }
```

index.json (all players, small rows)

```
[ { id, name, team, team_id, league,
position, position_group, nationality,
age, market_value_eur,
current_season: {apps, goals, assists, minutes},
last_season: {…}, last_season_name,
career: {apps, goals, assists} }, … ]
```

Enough for the home page cards, search, filters and sorting; 1.2 MB for 2,707 players.

meta.json

```
{ built_at, current_season, players,
clubs, leagues: {…}, source }
```

Why split into files?

Home page downloads 1 small file, not 57 MB.
A player page downloads only that player (~10–50 KB).
The CDN caches each file close to the visitor.

Figure 5. Shape of the dataset. Green = career level, blue = per season per competition.

9.1 Example: Haaland, 2025/26, as stored

Competition	Type	Apps	Goals	Assists	Minutes	Shots	Dribbles
Premier League (England)	league	35	27	8	2,958	126	17
FA Cup (England)	domestic_cup	4	3	1	231	6	1
EFL Cup	domestic_cup	3	0	0	199	4	2
Champions League	continental	10	8	0	755	35	1
FIFA Club World Cup	continental	4	3	1	255	15	2
World Cup Qualification UEFA	international	8	16	2	706	41	5
2025/2026 total		**64**	**57**	**12**	**5,104**	**227**	**28**

Premier League row cross-checked with Understat: 35 games, 27 goals, 8 assists, an exact match.

10 Pipeline stage 3: verification

File: `backend/pipeline/verify_dataset.py`. "It looks right" is not verification. We check the data two ways and write the results to `verification_report.json`.

10.1 Internal checks (hard errors fail the run)

- Every season's totals equal the sum of its competitions.
- No negative appearances, goals or assists.
- Every player in the index has a file. (Separate check: all 2,707 files have every field the website needs.)
- Counted, not failed: where FotMob's two endpoints disagree on goals/assists (now explained by row notes).

Result: 0 hard errors.

10.2 Independent cross-check with Understat

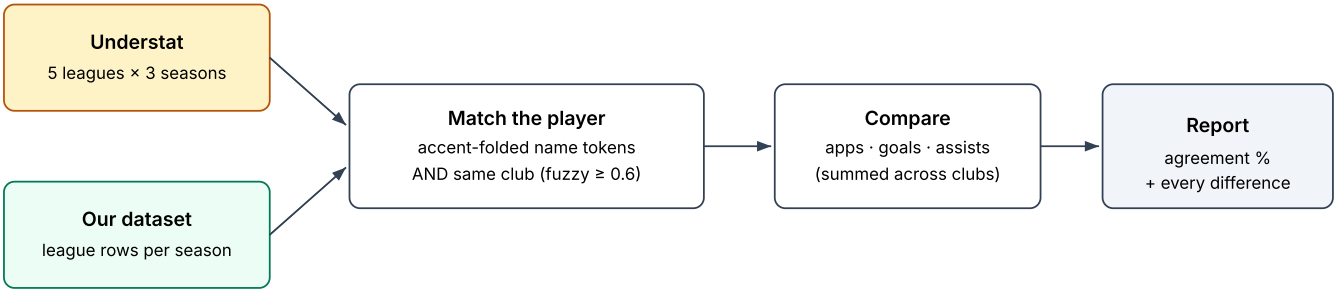


Figure 6. Cross-check algorithm.

Measure	Result
Player-league-seasons compared (2023/24, 2024/25, 2025/26)	4,734, of which 4,489 matched by name + club
Appearances agree	4,449 / 4,489 = **99.1%**
Goals agree	4,467 / 4,489 = **99.5%**
Assists agree	4,440 / 4,489 = **98.9%**

10.3 What the differences turned out to be

- **±1 assist** (Saka 9 vs 8, Damsgaard 10 vs 9): the two sources define assists differently. Opta's definition (FotMob) is the one leagues publish.
- **Big gaps at first** (Vitinha PSG vs Vitinha Marseille, two "Toni Martínez"): our matcher paired different people with the same name. Fix: require the same club as well. Big gaps went from 7 to 1.
- **The last one** (Amad Diallo 2023/24: 9 apps vs 1): Understat's record is incomplete; 9 is correct.

> **Lesson**
> Verification can be wrong too. When a check fails, investigate whether the data or the check is at fault. Two of our "data errors" were actually bugs in the verifier and in the builder.

11 The frontend: how the website works

11.1 Next.js App Router in 60 seconds

In Next.js, **folders are URLs**. `src/app/page.tsx` is `/`; `src/app/players/[id]/page.tsx` is `/players/anything`, with the square brackets meaning "a variable part of the URL". Files in `public/` are served as they are, so `public/data/index.json` is available at `/data/index.json`.

Components are **server components** by default (rendered on the server/at build time). A file starting with `"use client"` runs in the browser and can use state, clicks and effects. We use:

File	Kind	Job
`app/layout.tsx`	server	HTML skeleton, Geist font, page title "Football Transfer Market".
`app/page.tsx`	client	Home: loads index + meta, search box, league/position/club filters, sort, cards, "Show more".
`app/players/[id]/page.tsx`	server	Reads the `id` from the URL (in Next.js 16 `params` is a Promise: `await params`) and renders PlayerView.
`app/players/[id]/PlayerView.tsx`	client	Fetches the player file; header, KPI cards, two tabs, tables, expandable detail grids.
`lib/types.ts`	types	Shape of every JSON file (the contract).
`lib/format.ts`	helpers	`n()` number formatting with "–" for null, `money()`, image URLs, labels, detailed-stat groups.
`lib/Img.tsx`	client	Image that hides itself when the photo/logo doesn't exist (no broken-image icons).

11.2 Home page logic

```
const [rows, setRows]   = useState<IndexRow[] | null>(null);   // data
const [query, setQuery] = useState("");                         // UI state
const [league, setLeague] = useState("All");  // … position, club, sort, limit

useEffect(() => {                        // runs once after first render
  fetch("/data/index.json").then(r => r.json()).then(setRows);
}, []);

const filtered = useMemo(() => {         // recomputed only when inputs change
  return rows.filter(r => matchesLeague && matchesPosition && matchesClub && matchesSearch)
             .sort(bySelectedKey);
}, [rows, query, league, position, club, sort]);

filtered.slice(0, limit).map(r => <Link href={`/players/${r.id}`}>…card…</Link>)
```

- **Why filter in the browser?** 2,707 rows is small; filtering in memory is instant and needs no server.
- **Accent-insensitive search:** `s.normalize("NFD").replace(/[̀-ͯ]/g, "")` makes "mbappe" find "Mbappé".
- **"Show more" (60 at a time):** rendering 2,707 cards with photos at once would be slow.

11.3 Player page logic

- **Career tab:** `career_by_competition` grouped by type (League → Domestic cup → Continental → National team), plus an "All competitions" total row.
- **Season tab:** one collapsible card per season (summary line: apps, goals, assists, minutes); opening it shows a table with a row per competition, the per-club split for transfers, and amber notes for coverage gaps.
- **One table component for both** (`StatTable`): columns Apps, Min, Goals, Assists, xG, xA, Shots, Dribbles, Chances, Tackles, Rating. Clicking a row with ▼ opens `DetailGrid`: all ~50 stats grouped into Playing time, Attacking, Creating, Carrying & duels, Defending, Goalkeeping, Team while on pitch, Discipline.
- **Honest display:** null → "–"; incomplete totals → "*"; footnote explaining which seasons have detailed stats; a caption on the career tab saying the detailed columns start in 2023/24; goalkeepers told minutes aren't published.
- **Edge cases handled:** invalid id → "Player not found"; 45 players with no senior games → a clear message instead of an empty table.

11.4 Styling

Tailwind classes directly in the markup, dark theme (`bg-slate-950` background, emerald accent), responsive grids (`sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4`), and wide tables that scroll sideways on phones with the competition column sticky (`sticky left-0`).

12 Hosting and deployment

12.1 Git and GitHub

Branch	What it holds
`main`	README + LICENSE (initial commit).
`claude/import-deployed-app`	The first site, downloaded from the old Vercel deployment.
`project-material`	Everything your other agent pushed (old backend, data, handoff docs).
`claude/data-pipeline-rebuild`	The rebuild (same content as v2; older commits show "Unverified").
`claude/data-pipeline-v2`	**Current version**: pipeline + dataset + new site, all commits verified.

What is *not* committed (`.gitignore`): `node_modules/` , `.next/` , `.vercel/` , `.env*` (secrets), the raw `cache/` (110 MB, re-creatable) and logs.

12.2 Vercel: preview → production

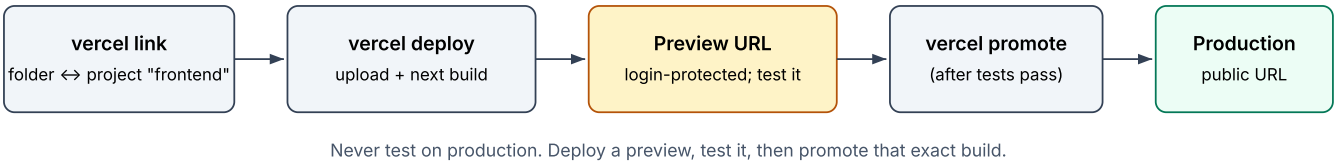


Figure 7. Deployment flow used for the live site.

- **Build output:** `/` is pre-rendered static HTML; `/players/[id]` is rendered on demand; all JSON files are static assets on the CDN.
- **Preview protection:** preview URLs redirect (302) to a Vercel login, so we tested them with `vercel curl` , which authenticates as you.
- **Why it works with your laptop off:** nothing points to your machine anymore; everything lives on Vercel.
- **History:** three deploy attempts from your machine six days earlier had failed (status Error); production had stayed on a 12-day-old build until our promotion.

12.3 Testing before handing over

Test	Tool	Result
Lint (code-quality rules)	ESLint	0 problems
Type check + production build	TypeScript, `next build`	Compiled successfully
Shape of all 2,707 player files	Python script	All fields present; found 45 players with no seasons → fixed UI
Browser sweep (home interactions + 150 player pages, clicking tabs and rows)	Playwright / Chromium	150/150, 0 JS errors, 0 failed requests (local and live)
Every player file on the live site	curl × 2,707 (16 in parallel)	2,707 × HTTP 200

13 End-to-end: opening a player page

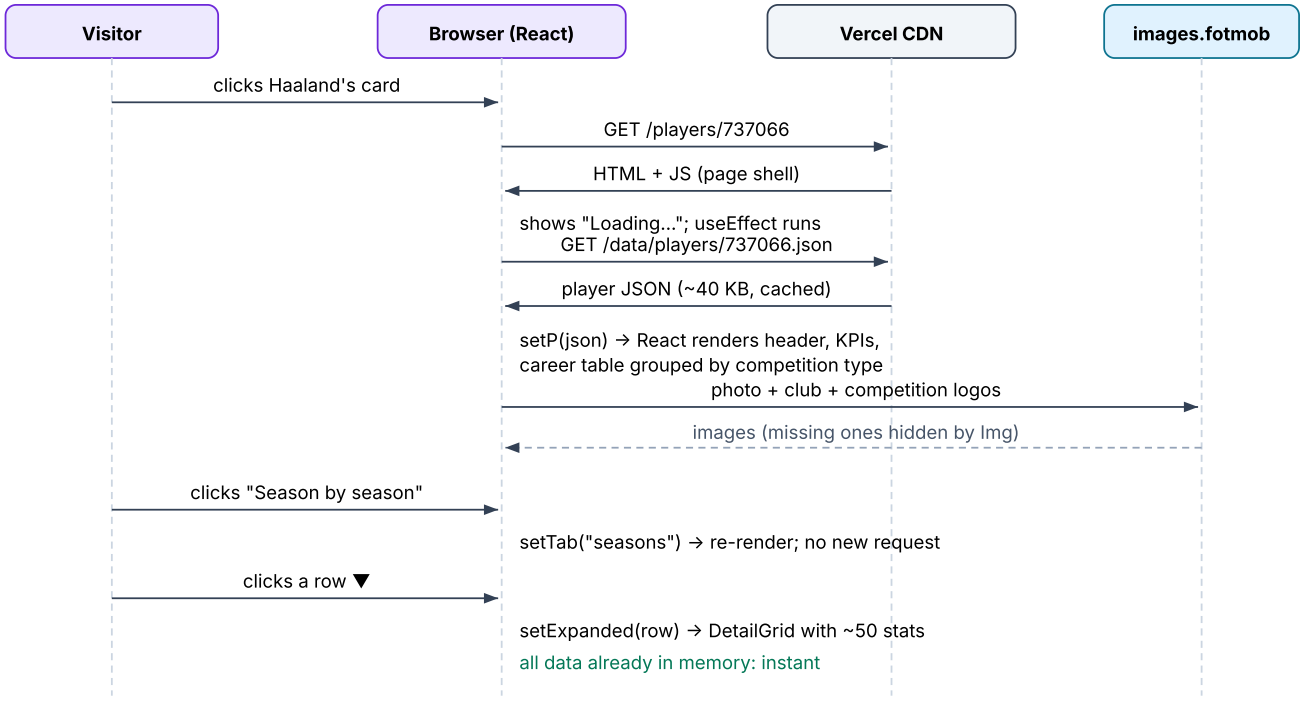


Figure 8. Sequence of events when a visitor opens a player and explores it.

Note what is **absent**: no database query, no API server, no Python. The heavy lifting was done by the pipeline hours earlier. That's why the page is fast and can't "go down" when a server or laptop is off.

13.1 The full data journey of one number

Haaland's 27 Premier League goals in 2025/26:

1. Opta records each goal at the match. FotMob shows Opta data.
2. `scrape_fotmob.py profiles` saves his profile: `careerHistory → senior → seasonEntries["2025/2026"] → tournamentStats[leagueId 47].goals = "27"`.
3. `scrape_fotmob.py deep` saves `cache/deep/737066/2025-2026_47.json` with goals 27, xG 25.81, shots 126, minutes 2,958....
4. `build_dataset.py` parses "27" → 27, buckets it into 2025/2026, labels it "Premier League (England)", type league, attaches the detailed stats, adds it to the season total (57) and to his career Premier League total (121).
5. `verify_dataset.py` fetches Understat EPL 2025: Erling Haaland, Manchester City, 27 goals. Match.
6. Committed in `public/data/players/737066.json`, deployed to Vercel.
7. The browser fetches the file and `n(27)` prints "27" in the Goals column.

14 Problems we hit and how we solved them

These are the real problems from this project, in the order they happened. Reading this table is probably the fastest way to learn what goes wrong in data projects.

#	Problem	How we noticed	Fix / how it's avoided
1	Old stats were projections	Read the scraper code	Rebuilt from a source with real current data; rule "never estimate".
2	Main stats sites block bots	403/405 reachability test	Chose sources that serve normal requests; no bypassing.
3	Network policy blocked vercel.com at first	Fetch failed (ENOTFOUND / 403)	Allowed the domain in environment settings.
4	Scraping too slow (0.7/s)	Progress log	10 threads + shared rate lock; resumed thanks to cache.
5	Same league name in different countries	Haaland showed "Bundesliga" twice	Country from FotMob league list appended when ambiguous.
6	One unknown value wiped whole totals	Career assists showed "unknown"	Sum of known values + "incomplete" flag + "*".
7	Norwegian cup classified as league	Example output	Extended cup name patterns.
8	Fake seasons "2024/2026", "2020/2022"	Screenshot	Multi-year national labels bucketed into their first season.
9	Team stats shown as goalkeeping stats for a striker	Screenshot	New group "Team while on pitch".
10	Goalkeeper field names guessed wrong	Inspected a keeper file	Mapped real ids (penalty_saves, ...). Lesson: inspect real data, don't guess.
11	React lint error: setState directly in useEffect	ESLint	Computed "invalid id" during render instead.
12	`PageProps` type "not found"	tsc	Next.js generates these types during build; ran `next build`.
13	Verifier matched wrong people with same name	Huge differences (Vitinha)	Match on name AND club.
14	Scraper crashed at 99.9% on `null` responses	Traceback in log	Null-safe trimming; rerun fetched only the 25 missing.
15	Mid-season transfers double-counted detailed stats	447 internal mismatches; Understat sided with combined numbers 79/82	Merge same-competition rows per season; keep per-club split.
16	Qualifiers not in detailed stats	Remaining mismatches analysed	Keep full numbers, add a visible note on the row.
17	Career minutes looked wrong (246 apps, 2,158 min)	Screenshot (Mbappé)	Caption: detailed columns cover 2023/24 onwards.

18	No minutes for goalkeepers	Inspected keeper data	Left as "–" (no estimating) and explained on the page.
19	Local test server served a stale build	404/500 in tests	Stop old server before restarting; test against a fresh build.
20	Broken-image icons for missing photos	Pre-handover review	`Img` component hides failed images.
21	45 players with no senior games → empty tables	Shape check of all files	Clear "no appearances yet" message.
22	Partial data almost committed to Git	Git hook warning	Moved partial output out; committed only the verified final build.
23	A GitHub token was pasted into chat	Visible in conversation	Token revoked immediately; tokens belong in environment variables, never in chat or code.
24	Commits shown as "Unverified"	Git hook	Fresh branch `claude/data-pipeline-v2` with verified identity (no history rewrite needed).

15 Other folders in the repository

The `project-material/` folder also contains parts of the first version. They are **not used by the live site**, but they're worth understanding because they show a different ("bigger") architecture.

Folder	Language / tech	What it is	Status
`backend/main.py`, `models.py`, `schemas.py`, `database.py`	Python, FastAPI, SQLAlchemy, Pydantic, SQLite	REST API (`/players`, `/players/{id}`, `/stats/summary`) over a SQLite DB with tables `players` and `player_competition_stats`.	replaced by static JSON
`backend/*.py` (~55 scripts)	Python	Earlier data fixes: enrich, dedupe, "brutal" verification loops, valuation.	legacy
`backend/sql_app.db`, `real_data.csv`, `raw_data_*.txt`	SQLite / CSV / text	Old dataset.	legacy
`ingestion/ingest.py`	Python, requests, pandas, boto3	API-Football → AWS S3 loader.	never run in production
`ml_model/train.py`	Python, XGBoost, scikit-learn, pandas	Market-value model trained from a Postgres feature table.	not used
`dbt/`	dbt (SQL transformations)	Project config only; no models written.	scaffold
`terraform/`	Terraform (HCL), AWS	Infrastructure as code: ECS cluster, RDS Postgres, load balancer, S3 bucket.	never deployed

When would you use that bigger architecture?
When data changes every few seconds (live scores), users write data (accounts, comments), or the dataset is far too big to ship as files. For read-only stats refreshed daily, it adds cost and failure points without benefit. Start simple; add a database and API only when a requirement forces you to.

16 How to run, update and extend it

16.1 Refresh the data (e.g. weekly)

```
git clone https://github.com/ssupreethsg/football-transfer-market.git
cd football-transfer-market && git checkout claude/data-pipeline-v2
cd project-material/backend/pipeline
pip install requests

rm -rf cache/players cache/deep cache/teams cache/leagues   # force fresh downloads
python scrape_fotmob.py squads
python scrape_fotmob.py profiles --workers 10
python scrape_fotmob.py deep --workers 10 --seasons 4
python build_dataset.py
python verify_dataset.py            # must print "hard errors: 0"
```

16.2 Run the website locally

```
cd ../../frontend
npm ci                 # install exact dependency versions
npm run dev            # http://localhost:3000, live reload
npm run lint && npm run build   # what must pass before deploying
```

16.3 Deploy

```
vercel link --project frontend      # once
vercel deploy                       # preview; test it
vercel promote <preview-url>        # make that build live
```

16.4 Ideas for extending it

- **Automatic weekly refresh:** a GitHub Actions workflow that runs the pipeline, commits the new data, and Vercel redeploys.
- **More history:** `--seasons 8` for detailed stats further back (longer run).
- **Charts:** goals per season as a bar chart on the player page.
- **Compare two players** side by side (both files are already separate JSONs).
- **Transfers page:** FotMob league pages also contain transfers; a new pipeline stage + page.

17 Lessons for your next project

17.1 A planning template (copy this)

Question	This project's answer
1. What exactly must the user see/do?	Search players; see career + season stats split by competition.
2. Where does the data come from, and is it allowed and reachable?	FotMob (Opta), Understat; tested with curl first.
3. How often does it change?	After each match day → batch rebuild is fine.
4. Does anyone write data?	No → no database or backend needed.
5. What's the smallest end-to-end slice?	One player (Haaland), raw data to screen.
6. How will I know it's correct?	Independent source + internal invariants + browser tests.
7. Where will it run, and what if my laptop is off?	Vercel CDN; laptop irrelevant.

17.2 Data and pipeline checklist

- Look at real raw responses before writing transform code (several bugs came from guessing field names).
- Cache every request; make every stage resumable; write files atomically.
- **Unknown ≠ 0.** Keep `null`, show "–".
- Never let a fallback silently pretend to succeed.
- Watch for double counting when combining two views of the same data.
- Separate counts (sum) from rates (weighted average).
- Verify against an independent source, and verify the verifier.
- Build one example end to end before scaling up.

17.3 Frontend checklist

- Write the data shape as TypeScript types first; let the compiler check you.
- Keep files small: one big file (the old 947-line page) is hard to fix; we split into page, view, table, grid, helpers.
- Load only what a page needs (index vs one player file).
- Design every state: loading, error, not found, empty, missing image.
- Read your framework version's docs; Next.js 16 changed APIs (e.g. `params` is a Promise).
- Run lint, build and a real-browser click-through before every deploy.

17.4 Deployment and security checklist

- Deploy to preview, test, then promote. Never debug on production.
- Never paste tokens or passwords into chats, code or commits; use environment variables; revoke anything exposed.
- Use fine-grained tokens limited to one repository, with short expiry.
- Don't commit generated or partial data; commit the verified final output.
- Don't point production at `localhost`.

18 Glossary

Term	Meaning
API	A URL that returns data (usually JSON) for programs instead of a web page for people.
JSON	Text format for data: `{"name": "Haaland", "goals": 27}`.
Scraper	Program that downloads data from websites/APIs automatically.
Cache	Saved copy of a download so it doesn't need to be fetched again.
Rate limit	Maximum request speed, to be polite and avoid being blocked.
Pipeline	Chain of steps that turns raw input into finished output (collect → transform → verify).
Static site	A site whose files are prepared in advance; the server just hands them out.
CDN	Network of servers worldwide holding copies of your files near visitors (Vercel provides one).
Next.js	React framework adding routing, building and hosting conventions.
React component	Function returning UI; re-runs when its state changes.
useState / useEffect / useMemo	React hooks: store values / run code after render (e.g. fetch) / cache a computed result.
Server vs client component	Runs on the server (no interactivity) vs in the browser (`"use client"`, clicks and state).
TypeScript type	Description of a data shape the compiler checks your code against.
Tailwind	CSS written as small class names in the markup.
Lint	Automatic check for code mistakes and bad patterns.
Preview / production deployment	A private test copy of the site / the public live site.
Branch / commit / push	Line of work / saved snapshot / upload snapshots to GitHub.
Opta (Stats Perform)	Company that records match events; the data source behind FotMob and the leagues.
xG / xA	Expected goals / expected assists: chance quality, from 0 to 1 per shot or pass.
Null	"No value / unknown". Different from 0.

End of guide. Everything described here is in the repository on branch `claude/data-pipeline-v2`; the pipeline README (`project-material/backend/pipeline/README.md`) has the short version of how to run it.